CSP-203: Software Tools & Technologies Lab



Chapter – 7

Docker: Network, Swarm, Kubernetes & Debugger: GDB

(Lab Session – 23)

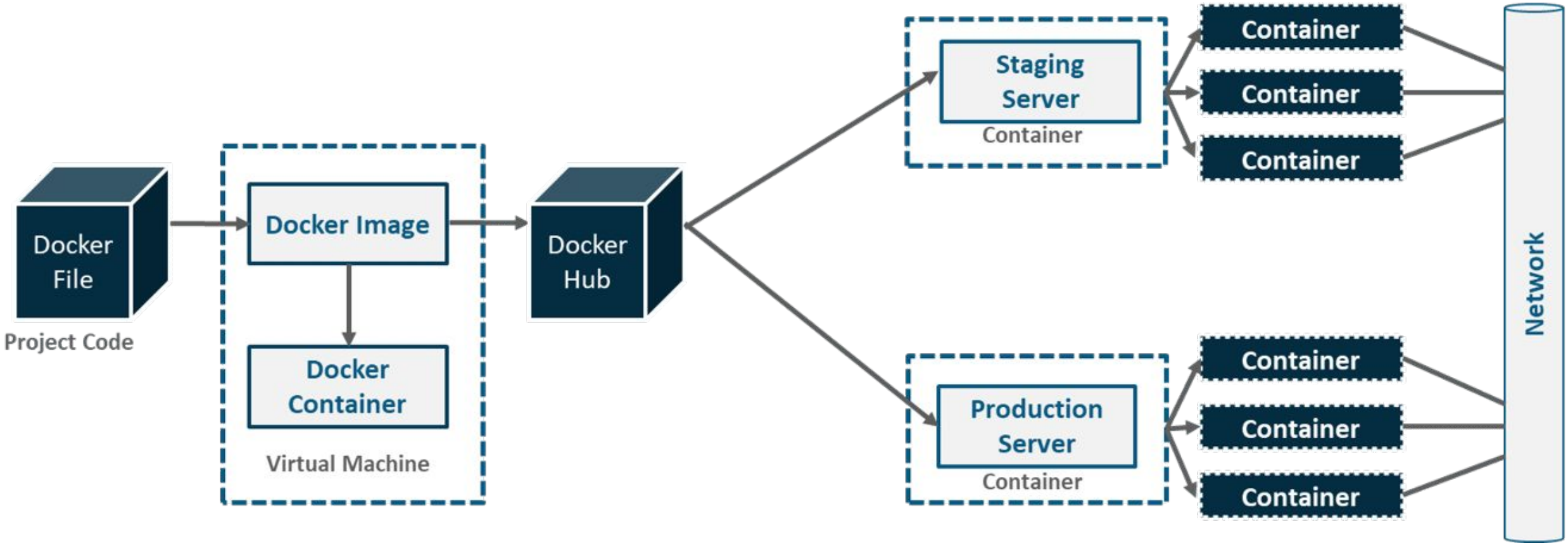
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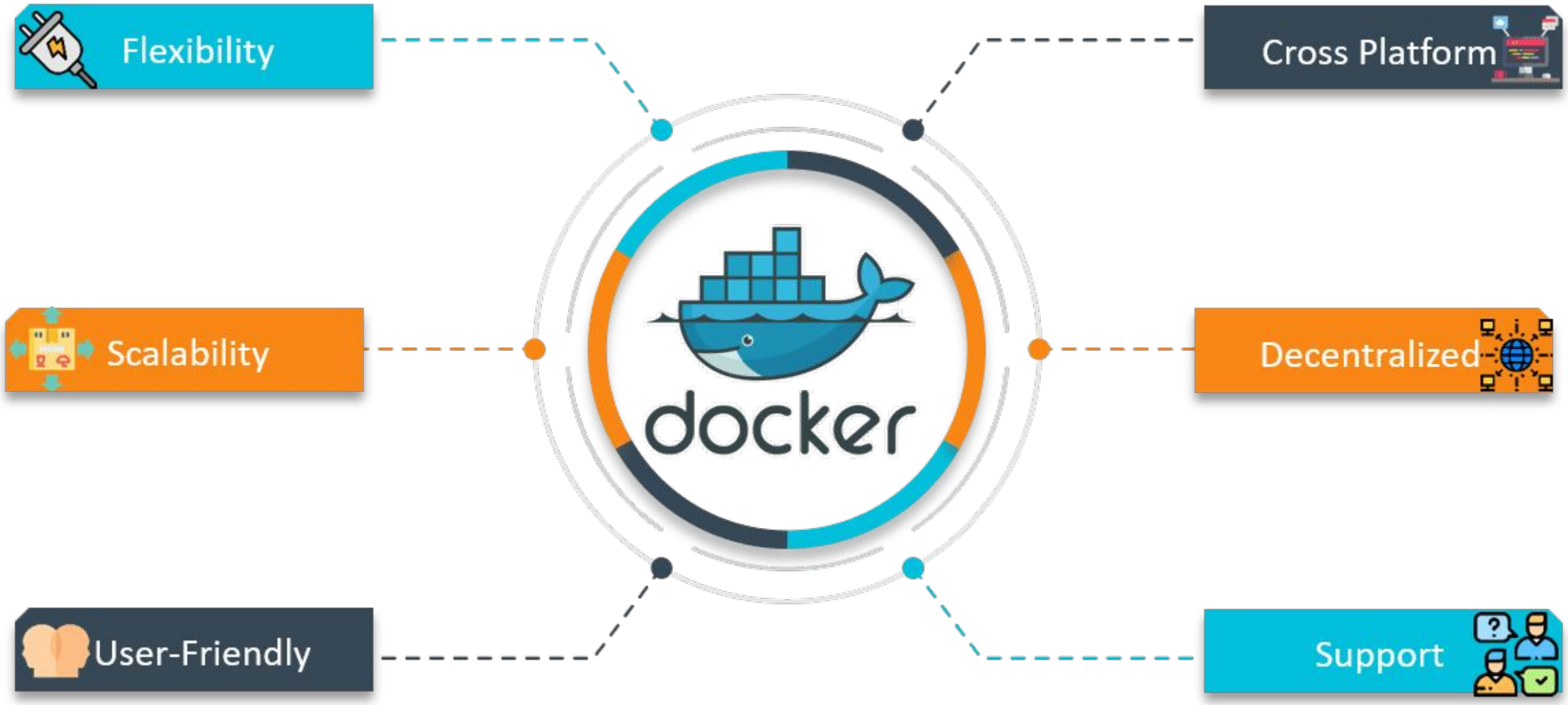
Contents

- ❑ Docker Networking
- ❑ Docker Swarm
- ❑ Docker Kubernetes
- ❑ GDB

Docker Networking

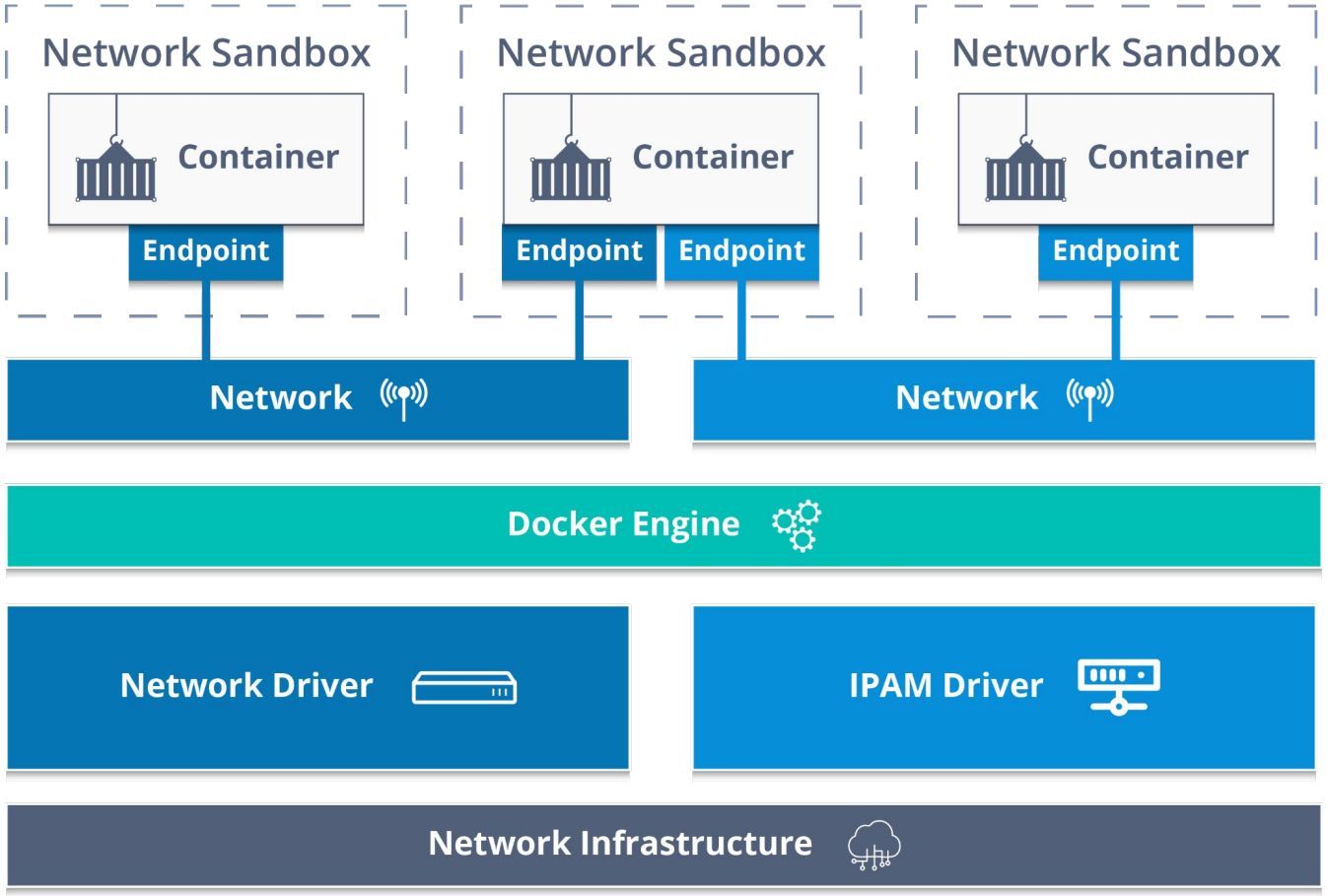


Goals of Docker Networking



Default Network Model

➢ Formalizes the steps required to provide networking for containers, while providing an abstraction that can be used to support multiple networks.

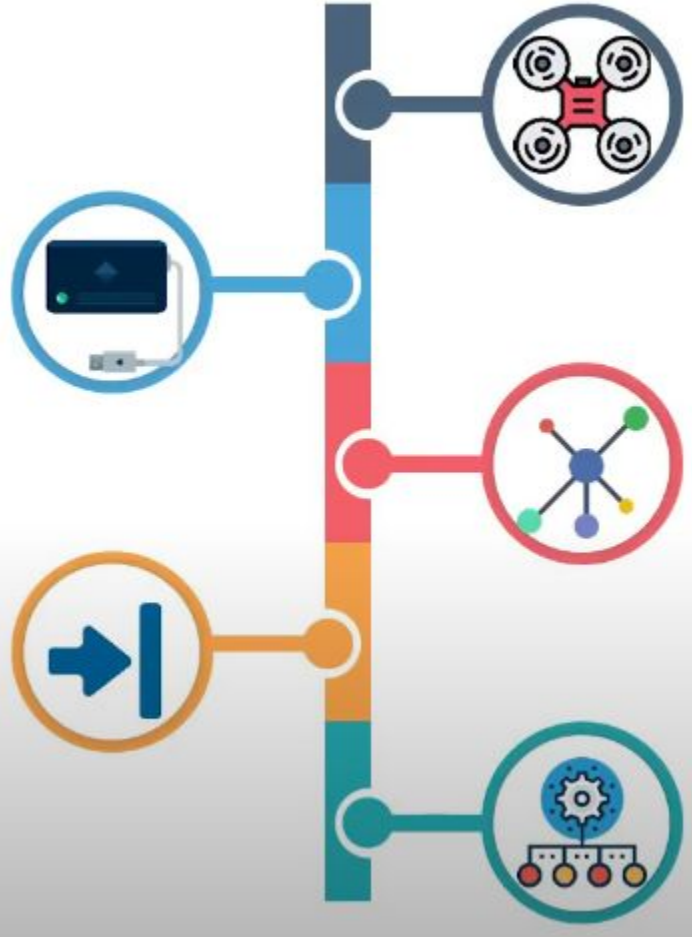


Architecture of Container Networking Model

Goals of Docker Networking

Network Controller
Provides the entry-point into libnetwork

Driver
Provides the actual implementation that makes network work



Network
Is an implementation of CNM : Network

End Point
Provides the connectivity for services exposed by a container in a network

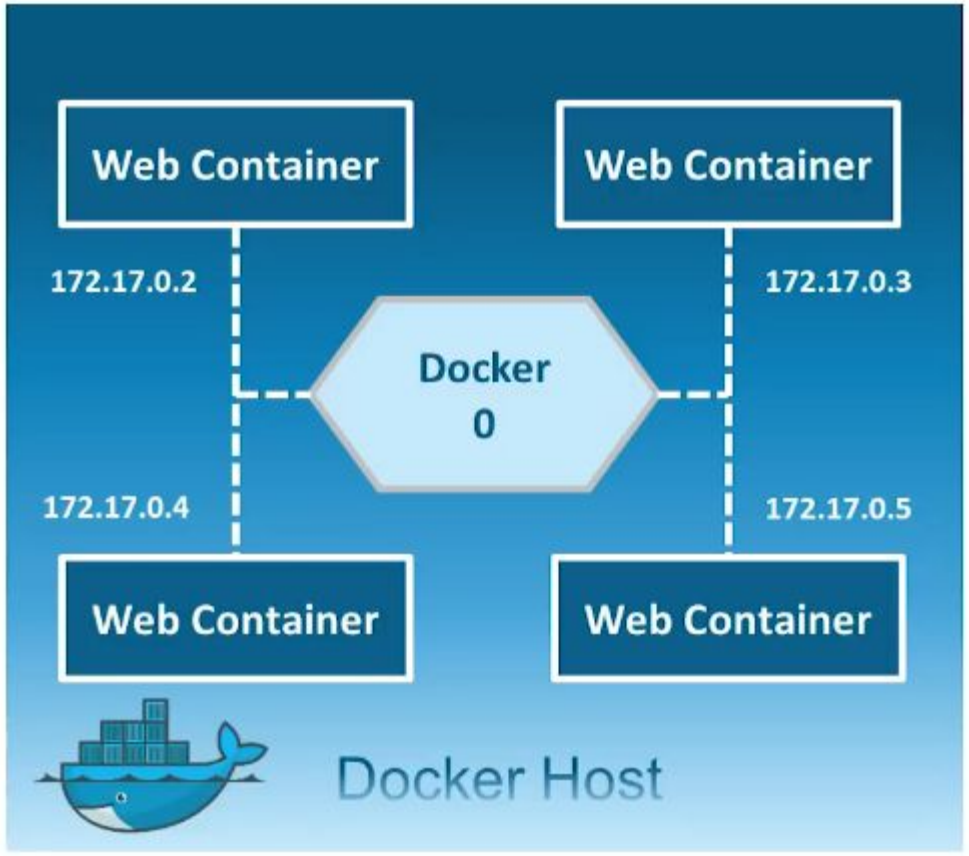
Sandbox
Represents container's network configuration

Network Drivers



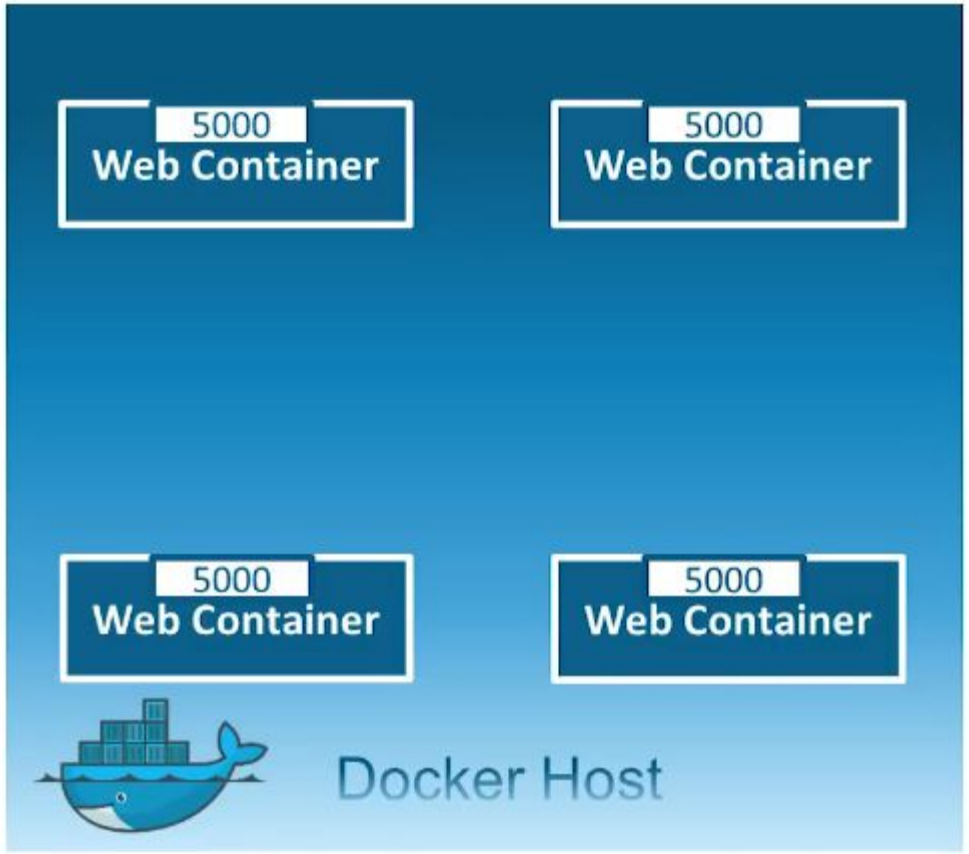
Bridge Network Driver

- Default Network Driver.
- If you don’t specify a driver, this type of driver you are creating.
- Used when applications run in standalone containers that need to be communicated.



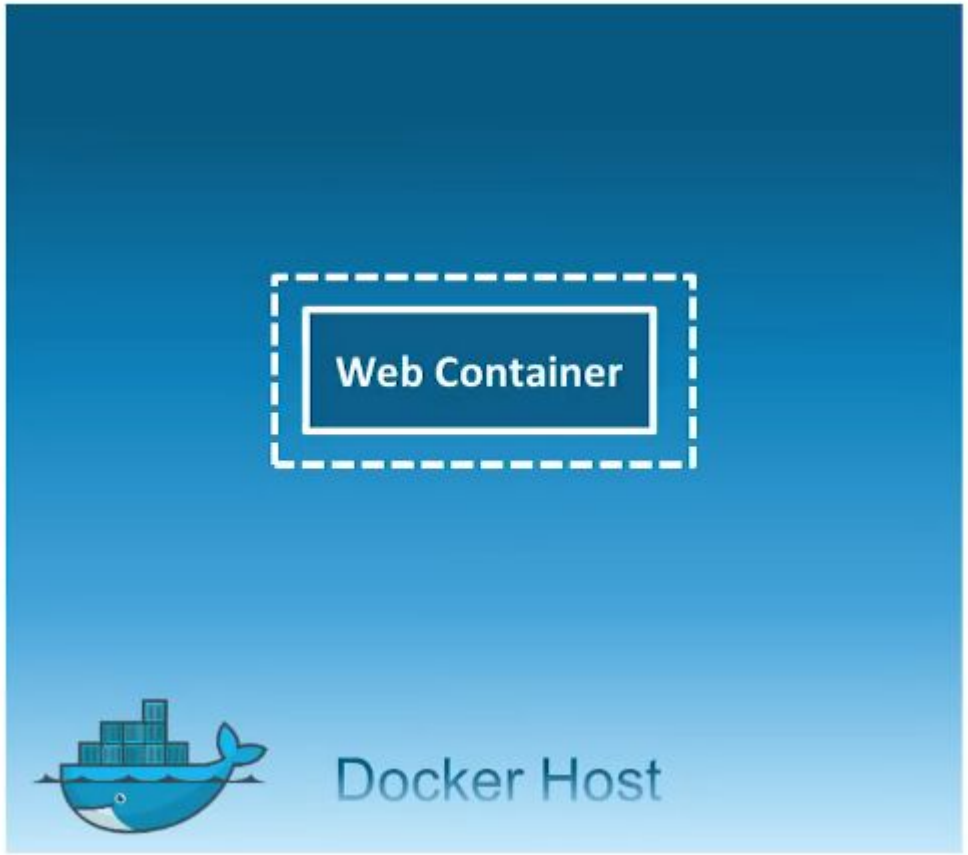
Host Network Driver

- For standalone containers, remove network isolation b/w the container and the docker hosts.
- Uses host's networking directly.



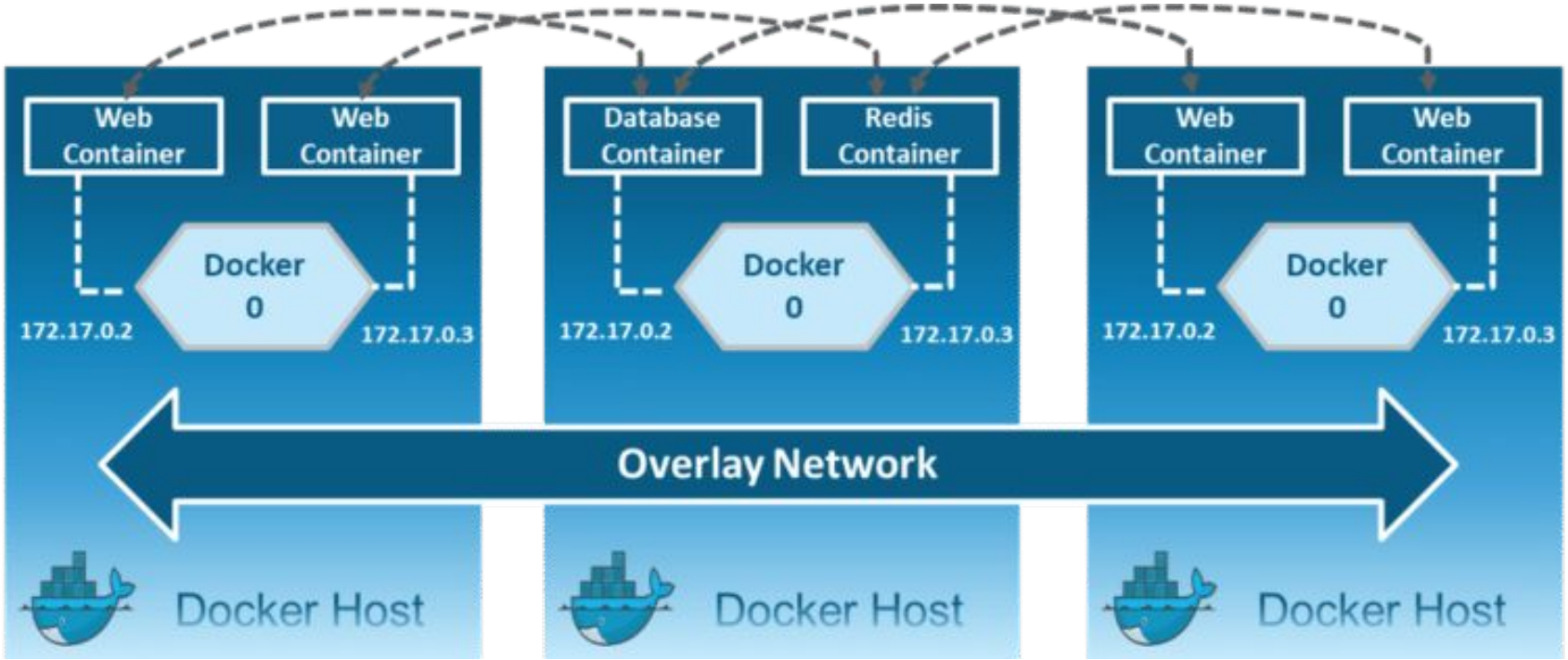
None Network Driver

- Disables all networking for containers.
- Used in conjunction with a custom network driver.



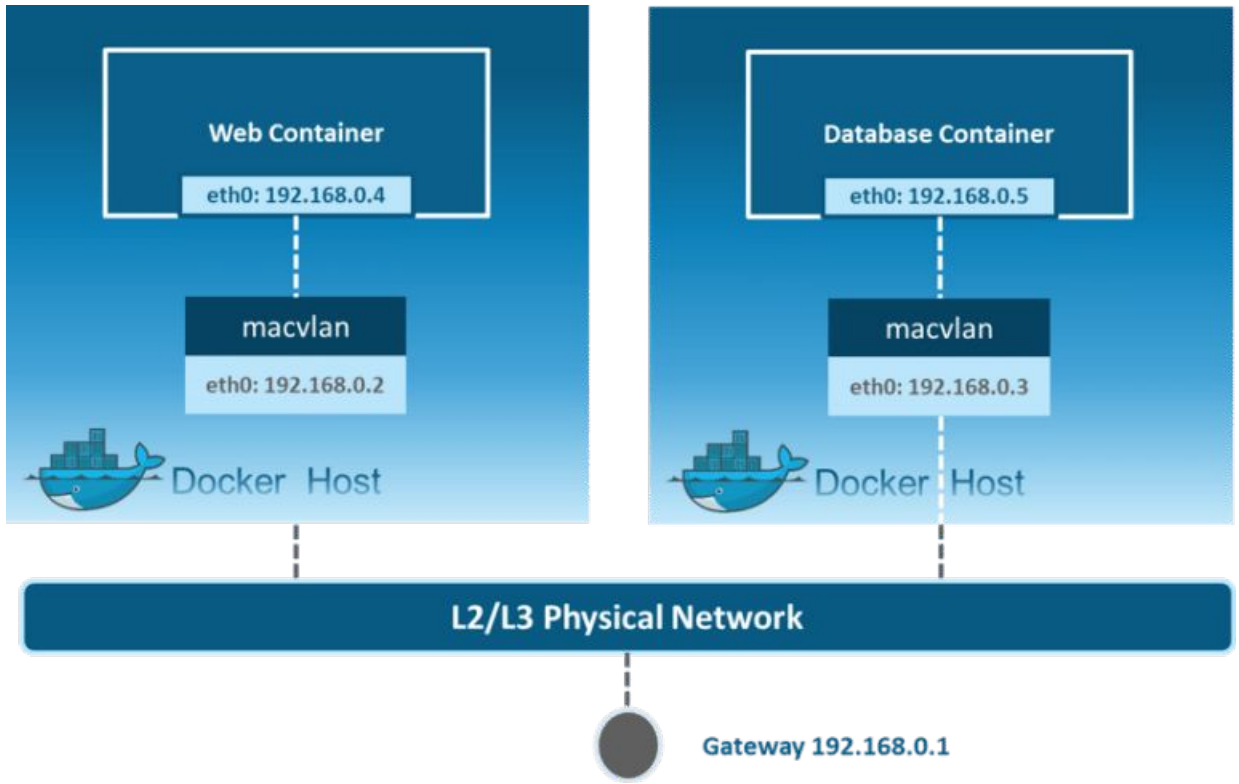
Overlay Networks

Connect multiple docker daemons together and enable swarm services to communicate with each other daemons.

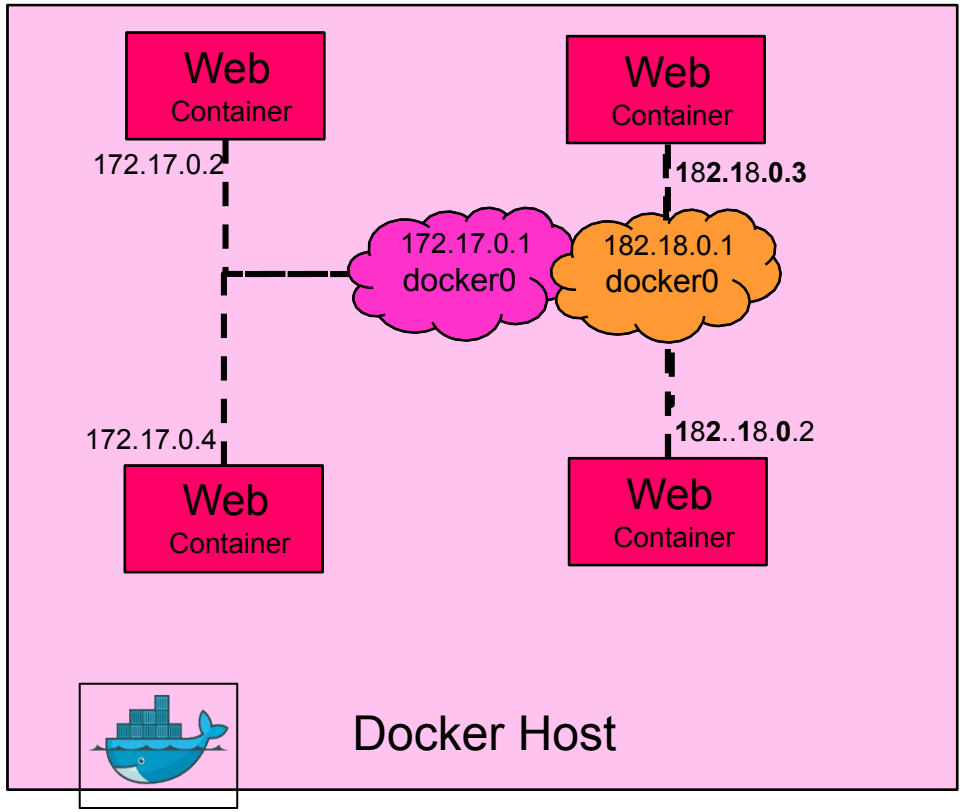


MacVlan

- Allows you to assign a MAC address to a container, making it appear as a physical device on your network.
- Route traffics to containers by their MAC addresses



User Defined Networks




```
docker network create \
    --driver bridge \
    --subnet 182.18.0.0/16
    --gateway 182.18.0.1
    custom-isolated-network
```

```
docker network ls
```

```
root@osboxes:/root # docker network ls
NETWORK ID          NAME                          DRIVER              SCOPE
dba0fb9370fe        bridge                        bridge              local
46d476b87cd9        customer-isolated-network     bridge              local
6de685cec1ce        docker_gwbridge               bridge              local
e29d188b4e47        host                          host                local
mmrho7vsb9rm        ingress                       overlay             swarm
d9f11695f0d6        none                          null                local
d371b4009142        simplewebappdocker_default    bridge              local
```

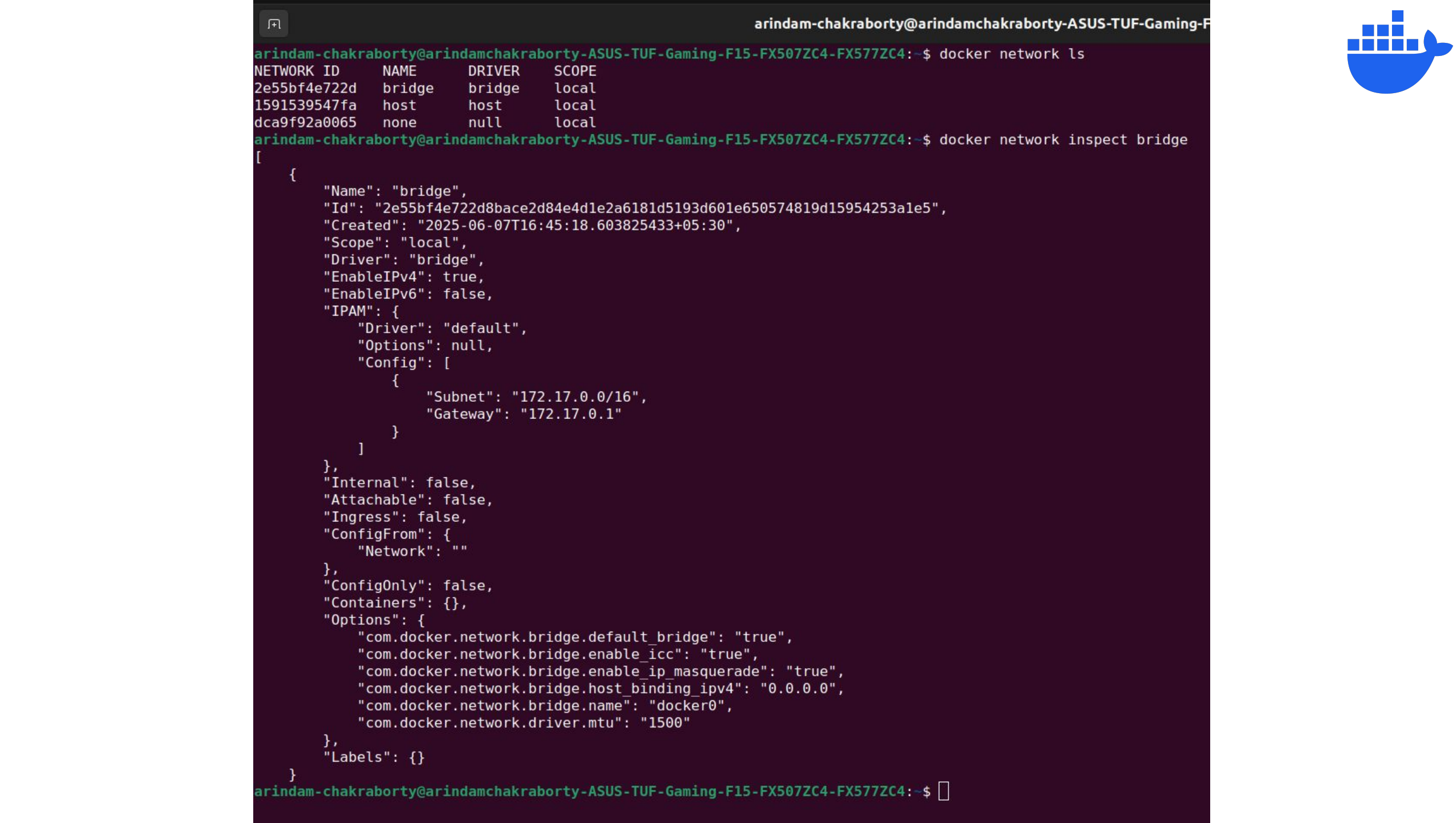
Network Inspect

```
docker inspect blissful_hopper
```

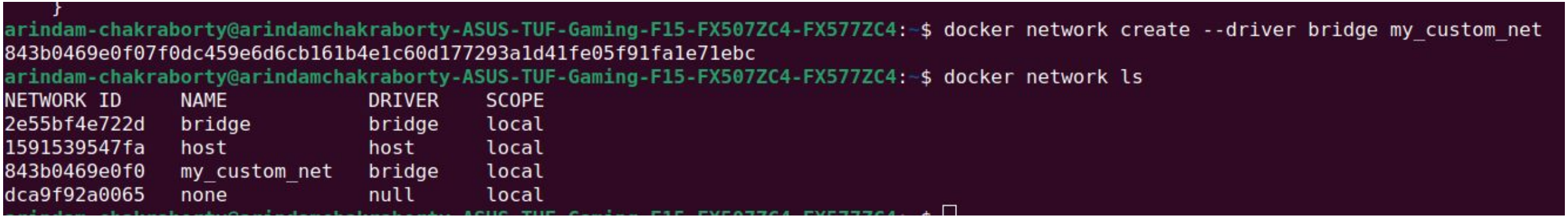

```
[
    {
        "Id":  "35505f7810d17291261a43391d4b6c0846594d415ce4f4d0a6ffbf9cc5109048",
        "Name": "/blissful_hopper",
        "NetworkSettings": {
            "Bridge": "",
            "Gateway": "172.17.0.1",
            "IPAddress": "172.17.0.6",
            "MacAddress": "02:42:ac:11:00:06",
            "Networks": {
                "bridge": {
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.6",
                    "MacAddress": "02:42:ac:11:00:06",
                }
            }
      }
]
```

Docker Default Networks

Network	Driver	Description
bridge	bridge	Default network for containers
host	host	Container shares host's network
none	null	No network for the container


arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker network ls
NETWORK ID NAME DRIVER SCOPE
2e55bf4e722d bridge bridge local
1591539547fa host host local
dca9f92a0065 none null local
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker network inspect bridge
[
{
"Name": "bridge",
"Id": "2e55bf4e722d8bace2d84e4d1e2a6181d5193d601e650574819d15954253a1e5",
"Created": "2025-06-07T16:45:18.603825433+05:30",
"Scope": "local",
"Driver": "bridge",
"EnableIPv4": true,
"EnableIPv6": false,
"IPAM": {
"Driver": "default",
"Options": null,
"Config": [
{
"Subnet": "172.17.0.0/16",
"Gateway": "172.17.0.1"
}
]
},
"Internal": false,
"Attachable": false,
"Ingress": false,
"ConfigFrom": {
"Network": ""
},
"ConfigOnly": false,
"Containers": {},
"Options": {
"com.docker.network.bridge.default_bridge": "true",
"com.docker.network.bridge.enable_icc": "true",
"com.docker.network.bridge.enable_ip_masquerade": "true",
"com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
"com.docker.network.bridge.name": "docker0",
"com.docker.network.driver.mtu": "1500"
},
"Labels": {}
}
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$

Creating Custom Networks



- Enables DNS-based communication
- Allows custom subnets and IP address assignment

Connect Containers to Networks

- At container creation:

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker run -dit --name web --network my_custom_net nginx
c52b7aad7e98083fd2edaa3efc6da769b3c48d3ead7688ff21819ae06dfcbb98
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS                     PORTS     NAMES
c52b7aad7e98   nginx     "/docker-entrypoint.…"   7 seconds ago   Up 7 seconds               80/tcp    web
96d970c2b15b   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               friendly_wescoff
cb4ab9f60cee   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               laughing_kowalevski
02a3e6b538ce   ubuntu    "echo 'Hello World'"     36 hours ago    Exited (0) 36 hours ago              myruncontainer
edd34874fe7f   ubuntu    "sleep 60"               36 hours ago    Exited (0) 36 hours ago              mycontainer
```

- After container creation: docker network connect my_custom_net web
- To Disconnect: docker network disconnect my_custom_net web
- If you want to assign your own ip with in that subnet then you have to use --ip <ip_address>
- Removing a network: docker network rm my_custom_net [Note: You have to stop and disconnect any active connections inside the network before removing that network]
- Installing ping: apt-get update && apt-get install iputils-ping
- Ping test for reachability to a node: ping <dest_container_name> / ping <dest_container_ip>

Inspecting Network Details

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker network disconnect my_custom_net web
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS                     PORTS     NAMES
c52b7aad7e98   nginx     "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes                         web
96d970c2b15b   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               friendly_wescoff
cb4ab9f60cee   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               laughing_kowalevski
02a3e6b538ce   ubuntu    "echo 'Hello World'"     36 hours ago    Exited (0) 36 hours ago              myruncontainer
edd34874fe7f   ubuntu    "sleep 60"               36 hours ago    Exited (0) 36 hours ago              mycontainer
4555487c5510   ubuntu    "/bin/bash"              36 hours ago    Exited (0) 35 hours ago              myubuntu
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker network inspect my_custom_net
[
    {
        "Name": "my_custom_net",
        "Id": "843b0469e0f07f0dc459e6d6cb161b4e1c60d177293a1d41fe05f91fa1e71ebc",
        "Created": "2025-06-07T22:32:40.776209431+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {}
    }
```

Connecting Two Networks

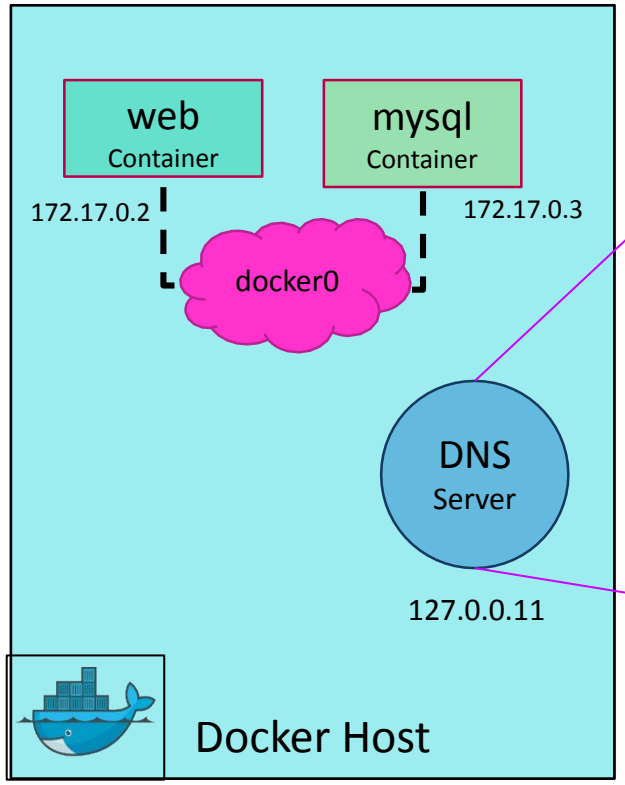
- To enable bridge forwarding: sudo iptables -P FORWARD ACCEPT

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker run -dit --name web --network my_custom_net nginx
c52b7aad7e98083fd2edaa3efc6da769b3c48d3ead7688ff21819ae06dfcbb98
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS                     PORTS     NAMES
c52b7aad7e98   nginx     "/docker-entrypoint.…"   7 seconds ago   Up 7 seconds               80/tcp    web
96d970c2b15b   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               friendly_wescoff
cb4ab9f60cee   ubuntu    "/bin/bash"              4 hours ago     Exited (0) 4 hours ago               laughing_kowalevski
02a3e6b538ce   ubuntu    "echo 'Hello World'"     36 hours ago    Exited (0) 36 hours ago              myruncontainer
edd34874fe7f   ubuntu    "sleep 60"               36 hours ago    Exited (0) 36 hours ago              mycontainer
```

- After container creation: docker network connect my_custom_net web
- To Disconnect: docker network disconnect my_custom_net web
- If you want to assign your own ip with in that subnet then you have to use --ip <ip_address>
- Removing a network: docker network rm my_custom_net [Note: You have to stop and disconnect any active connections inside the network before removing that network]
- Installing ping: apt-get update && apt-get install iputils-ping
- Ping test for reachability to a node: ping <dest_container_name> / ping <dest_container_ip>

Embedded DNS

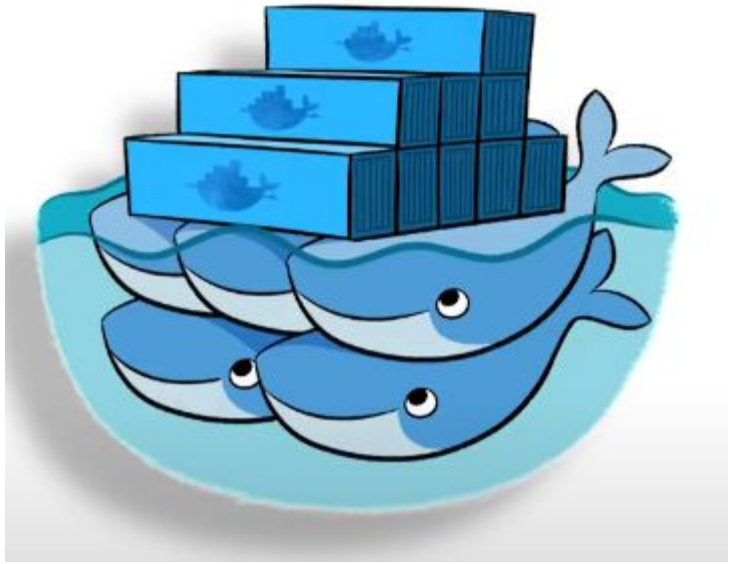
```
mysql.connect(    mysql    )
```



Host	IP
web	172.17.0.2
mysql	172.17.0.3

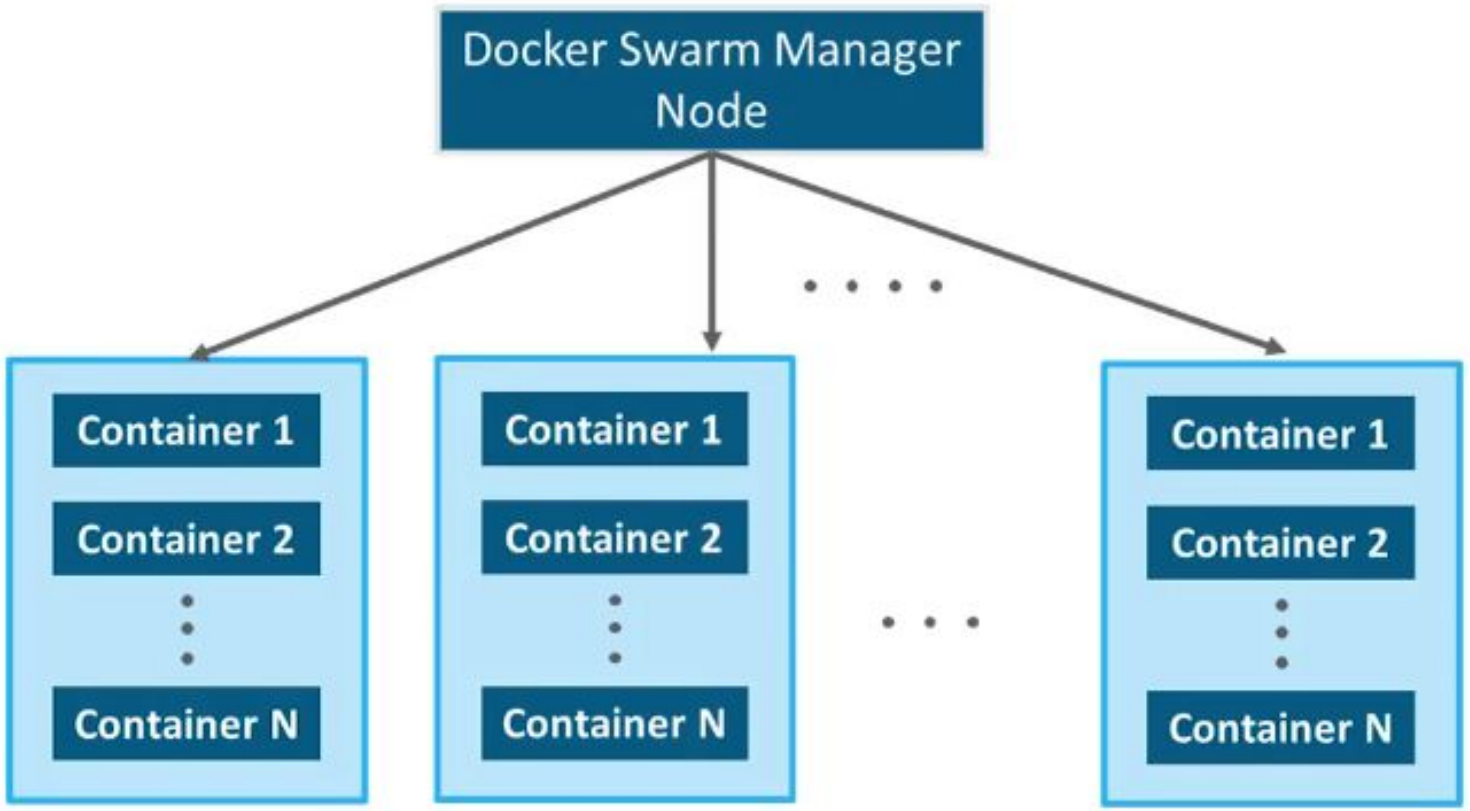
Docker Swarm

- Cluster of machines
- Provides scalable and reliable platform to run many containers.
- IT admin and dev can establish and manage cluster of docker nodes as a single virtual system.



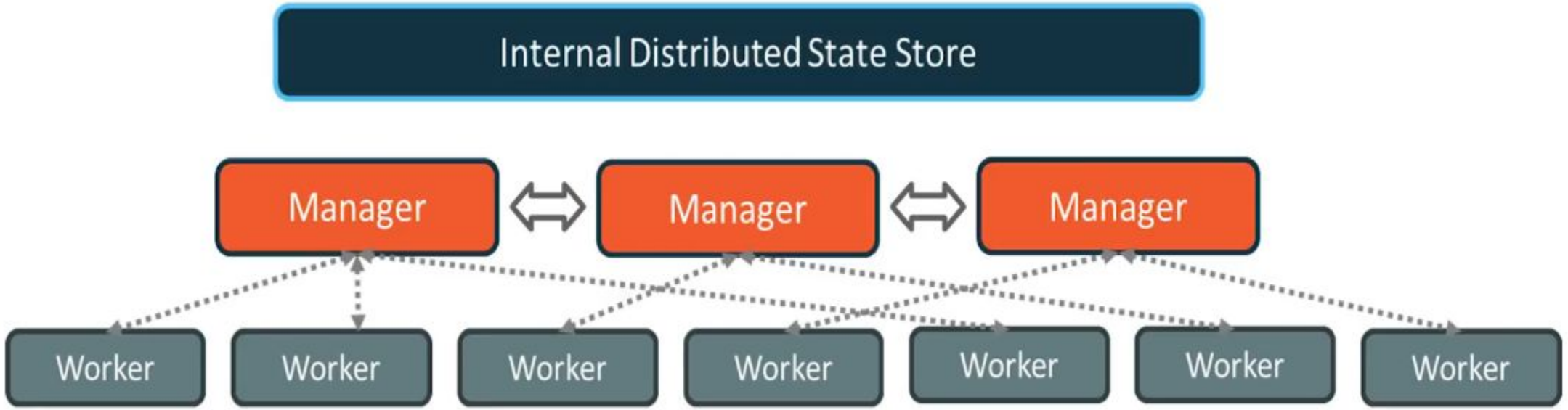
Docker Swarm Clusters

You can create swarm docker engines where you can deploy application services. So, you don’t need additional orchestration software to create or manage a swarm



Docker Swarm Managers and Workers

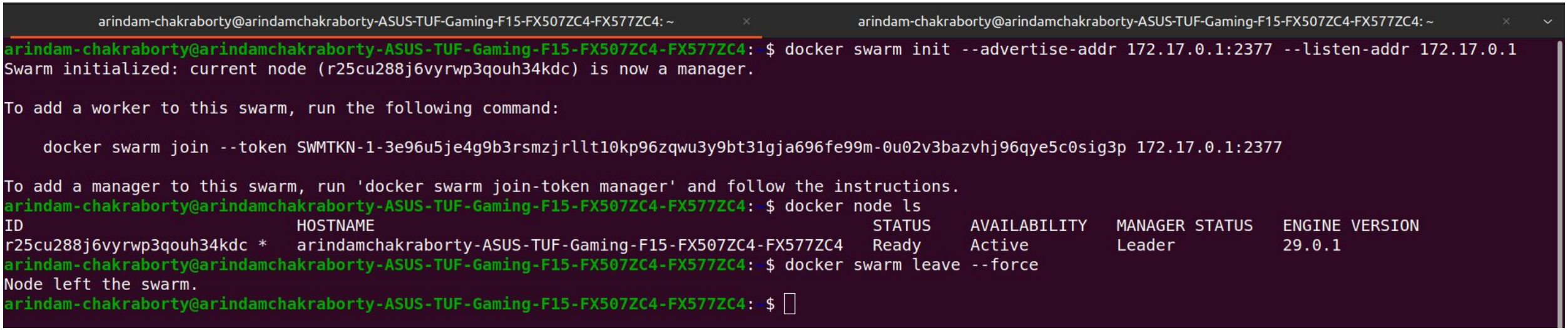
Every swarm has atleast one manager and has many worker nodes under it



Command Tables: Overview Docker Swarm Commands

Command	Description
docker swarm init	Initialize a swarm cluster
docker swarm join-token worker	Get token to join as worker
docker swarm join-token manager	Get token to join as manager
docker node ls	List nodes in the swarm
docker service create --name <name> --replicas <running_containers>	Deploy service in swarm
docker stack deploy -c	Deploy stack
docker service ps	View service tasks
docker service scale =	Scale service

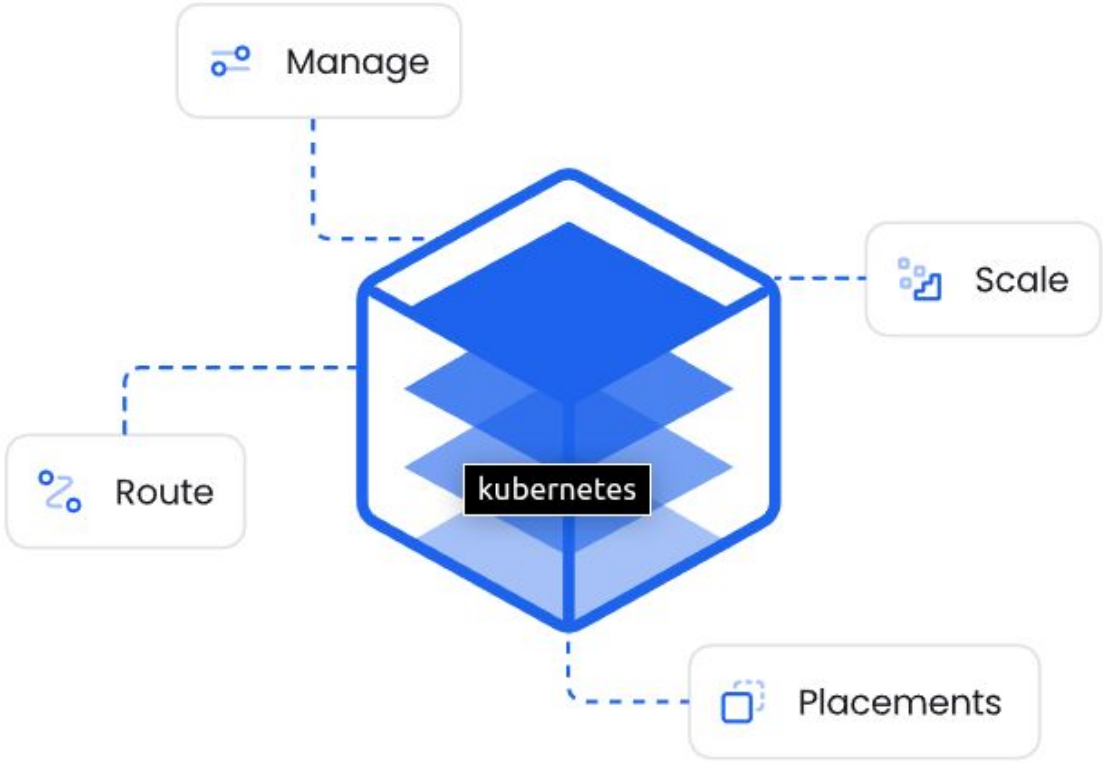
Command Tables: Overview Docker Swarm Commands



Docker Kubernetes

Introduction:

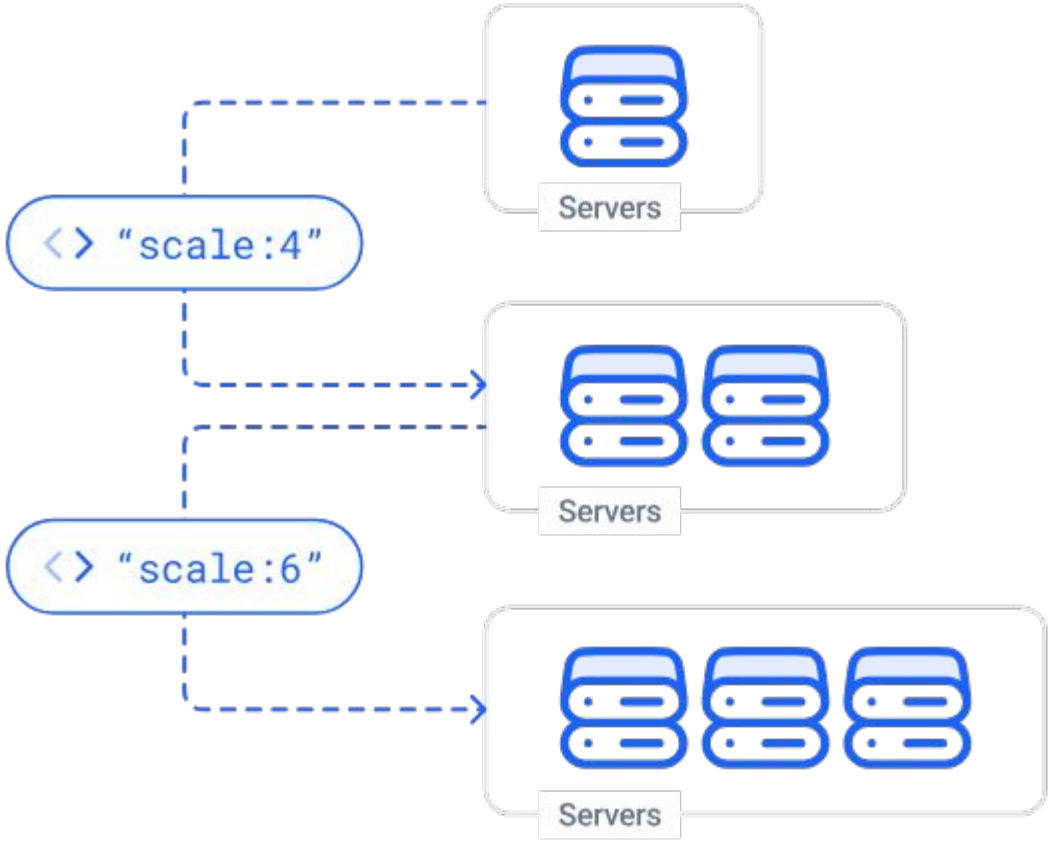
Kubernetes is an open source orchestration system for automating the management, placement, scaling, and routing of containers, gaining popularity among developers and IT operations teams. Docker Engine, the original container engine for Kubernetes, plays a pivotal role in managing container operations in the host environment. First developed by Google in 2014, Kubernetes (K8s) is now maintained by the Cloud Native Computing Foundation.



Docker Kubernetes

Advantages:

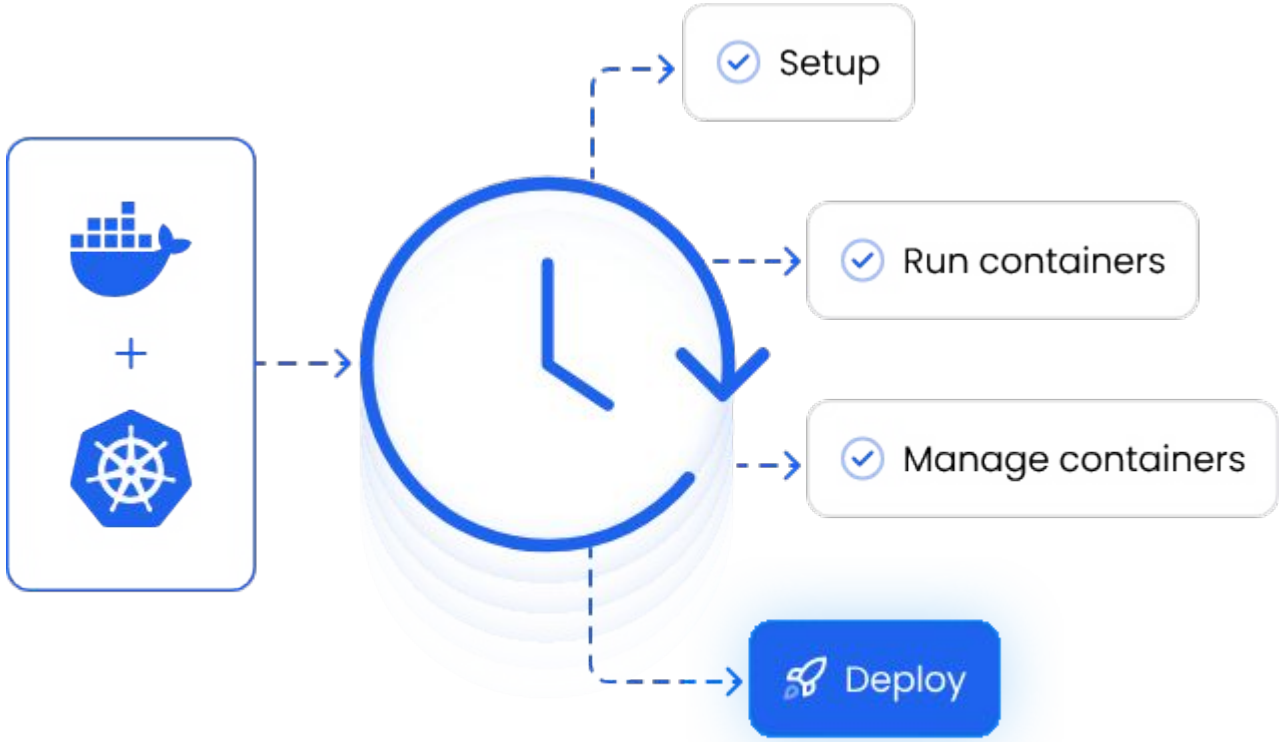
Kubernetes manages scaling and excels in availability, making it a popular choice among professional developers. The system is engineered to automatically adapt, scale, and distribute workloads efficiently, ensuring high availability even in large-scale operations.



Docker Kubernetes

Deploy in Minutes:

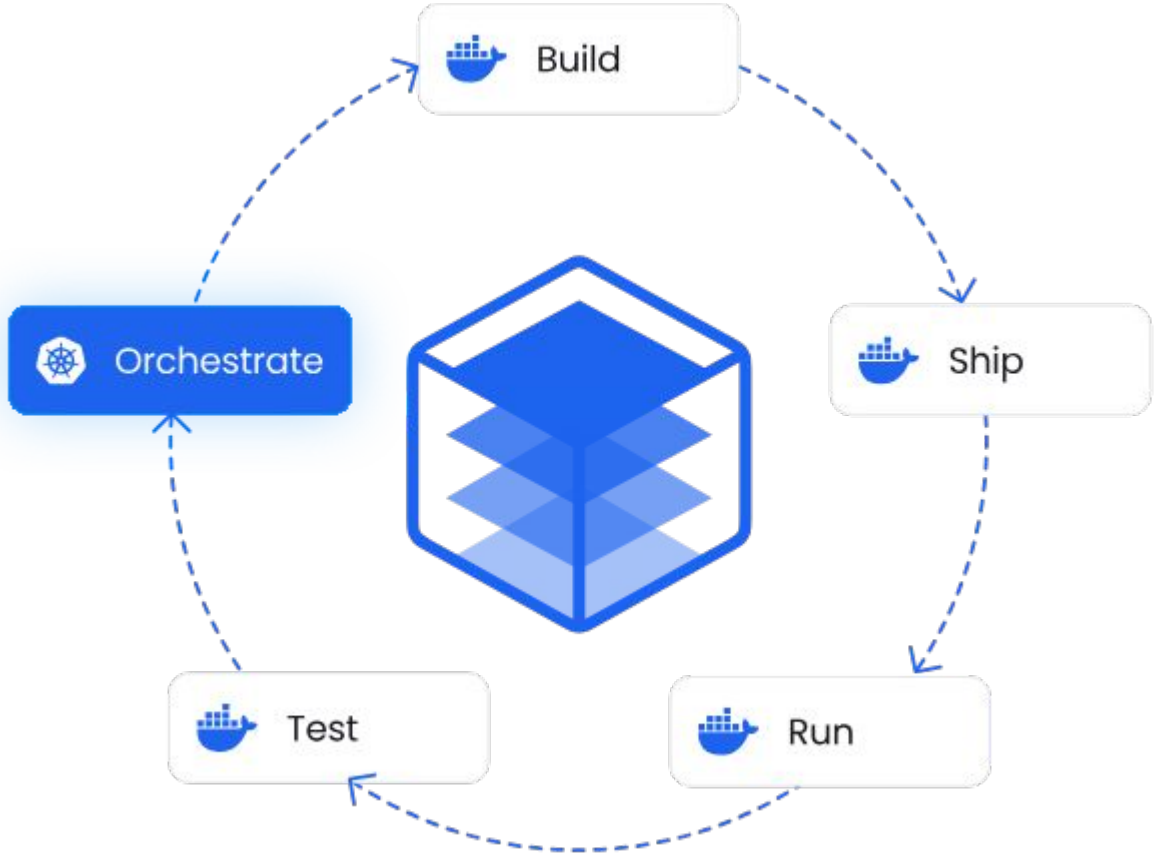
Docker Desktop simplifies the Kubernetes setup and installation process, saving time and reducing manual setup steps. With minimal user intervention, Docker can deploy a fully operational Kubernetes cluster within minutes.



Docker Kubernetes

Two strong tech with distinct usecase:

Kubernetes and Docker are a power couple when it comes to leveraging containers at larger scales. While Docker is a container runtime that lets developers build, ship, and run containers, Kubernetes offers container orchestration. Together, they make it possible to containerize applications and their systems at scale.



Command Tables: Overview Docker Kubernets Commands

Command	Description
kubectl get nodes	List cluster nodes
kubectl get pods	List running pods
kubectl get services	List services
kubectl apply -f .yaml	Create resources from YAML
kubectl delete -f .yaml	Delete resources from YAML
kubectl describe	Detailed resource info
kubectl logs	View logs of a pod
kubectl exec -it -- /bin/bash	Exec into a running pod

CSP-203: Software Tools & Technologies Lab



Chapter – 8

Debugger : GDB

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First Example: Area of a triangle (tarea.c)

Open tarea.c Save

```c
#include <stdio.h>
#include <stdlib.h>

int main ()
{
        int x1, y1, x2, y2, x3, y3;
        float area;

        printf("Program to calculate the area of a triangle with integer coordinates\n");

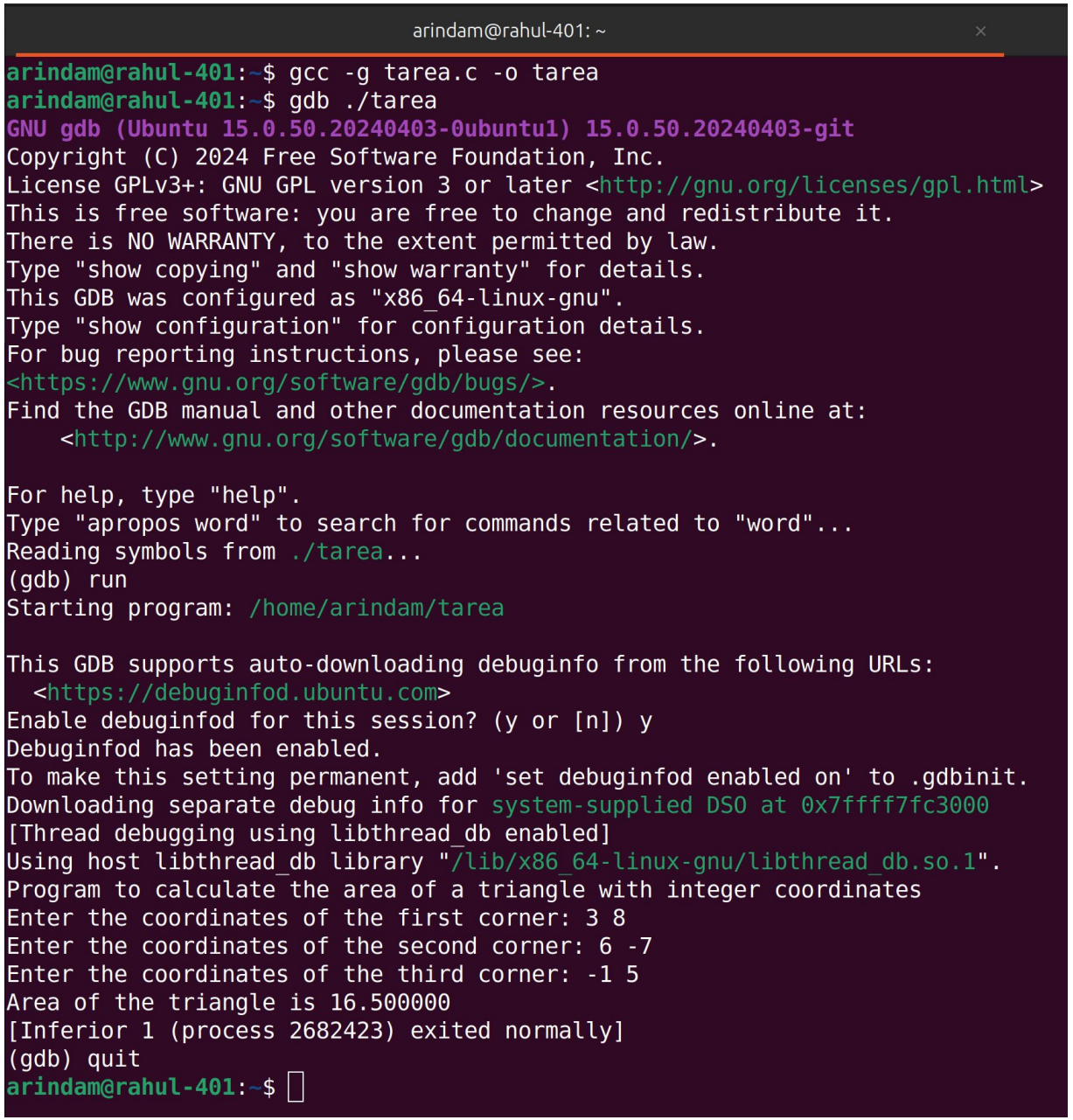
        printf("Enter the coordinates of the first corner: ");
        scanf("%d%d", &x1, &y1);
        printf("Enter the coordinates of the second corner: ");
        scanf("%d%d", &x2, &y2);
        printf("Enter the coordinates of the third corner: ");
        scanf("%d%d", &x3, &y3);

        area = x1 * (y2 - y3);
        area += x2 * (y1 - y3);
        area += x3 * (y1 - y2);
        if (area < 0) area = -area;
                area /= 2;

        printf("Area of the triangle is %lf\n", area);

        return 0;
}
```

Running Code with GDB



Listing Code:

arindam@rahul-401: ~

```
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) list
1       #include <stdio.h>
2       #include <stdlib.h>
3
4       int main ()
5       {
6               int x1, y1, x2, y2, x3, y3;
7               float area;
8
9               printf("Program to calculate the area of a triangle with intege
10
(gdb) list
11              printf("Enter the coordinates of the first corner: ");
12              scanf("%d%d", &x1, &y1);
13              printf("Enter the coordinates of the second corner: ");
14              scanf("%d%d", &x2, &y2);
15              printf("Enter the coordinates of the third corner: ");
16              scanf("%d%d", &x3, &y3);
17
18              area = x1 * (y2 - y3);
19              area += x2 * (y1 - y3);
20              area += x3 * (y1 - y2);
(gdb) list
21              if (area < 0) area = -area;
22                      area /= 2;
23
24              printf("Area of the triangle is %lf\n", area);
25
26              return 0;
27      }
28
(gdb) list -
11              printf("Enter the coordinates of the first corner: ");
12              scanf("%d%d", &x1, &y1);
13              printf("Enter the coordinates of the second corner: ");
14              scanf("%d%d", &x2, &y2);
15              printf("Enter the coordinates of the third corner: ");
16              scanf("%d%d", &x3, &y3);
17
18              area = x1 * (y2 - y3);
19              area += x2 * (y1 - y3);
20              area += x3 * (y1 - y2);
(gdb)
```

Listing Code:

```
(gdb) list 15
10
11              printf("Enter the coordinates of the first corner: ");
12              scanf("%d%d", &x1, &y1);
13              printf("Enter the coordinates of the second corner: ");
14              scanf("%d%d", &x2, &y2);
15              printf("Enter the coordinates of the third corner: ");
16              scanf("%d%d", &x3, &y3);
17
18              area = x1 * (y2 - y3);
19              area += x2 * (y1 - y3);
(gdb) list 18,22
18              area = x1 * (y2 - y3);
19              area += x2 * (y1 - y3);
20              area += x3 * (y1 - y2);
21              if (area < 0) area = -area;
22                      area /= 2;
(gdb)
```

Setting Breakpoint & Stepping Through:

```
For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break 11
Breakpoint 1 at 0x11d3: file tarea.c, line 11.
(gdb) run
Starting program: /home/arindam/tarea

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Program to calculate the area of a triangle with integer coordinates

Breakpoint 1, main () at tarea.c:11
11              printf("Enter the coordinates of the first corner: ");
(gdb) next
12              scanf("%d%d", &x1, &y1);
(gdb) next
Enter the coordinates of the first corner: 3 8
13              printf("Enter the coordinates of the second corner: ");
(gdb) next
14              scanf("%d%d", &x2, &y2);
(gdb) next
Enter the coordinates of the second corner: 6 -7
15              printf("Enter the coordinates of the third corner: ");
(gdb) next
16              scanf("%d%d", &x3, &y3);
(gdb) next
Enter the coordinates of the third corner: -1 5
18              area = x1 * (y2 - y3);
(gdb) next
19              area += x2 * (y1 - y3);
(gdb) next
20              area += x3 * (y1 - y2);
(gdb) continue
Continuing.
Area of the triangle is 16.500000
[Inferior 1 (process 2688927) exited normally]
(gdb)
```

Shortcuts about GDB command:

Command shortcuts: You do not have to type the entire command (like list or next). Certain abbreviations are allowed. Here is a short list.

- b break
- bt backtrace
- c continue
- d delete
- dis disable
- disp display
- do down
- en enable
- f frame
- fin finish
- h help
- i info
- l list
- n next
- p print
- q quit
- ret return
- r run
- s step

• Command name abbreviations are allowed if unambiguous.
• Hit return to repeat the last command.
• The unaltered source file must be present for gdb to run correctly.

More About Breakpoints:

```
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break main
Breakpoint 1 at 0x11b5: file tarea.c, line 5.
(gdb) break 11
Breakpoint 2 at 0x11d3: file tarea.c, line 11.
(gdb) break 18
Breakpoint 3 at 0x126c: file tarea.c, line 18.
(gdb) break 24
Breakpoint 4 at 0x1304: file tarea.c, line 24.
(gdb) info break
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x00000000000011b5 in main at tarea.c:5
2       breakpoint     keep y   0x00000000000011d3 in main at tarea.c:11
3       breakpoint     keep y   0x000000000000126c in main at tarea.c:18
4       breakpoint     keep y   0x0000000000001304 in main at tarea.c:24
(gdb) disable 3
(gdb) info break
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x00000000000011b5 in main at tarea.c:5
2       breakpoint     keep y   0x00000000000011d3 in main at tarea.c:11
3       breakpoint     keep n   0x000000000000126c in main at tarea.c:18
4       breakpoint     keep y   0x0000000000001304 in main at tarea.c:24
(gdb) enable 3
(gdb) info break
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x00000000000011b5 in main at tarea.c:5
2       breakpoint     keep y   0x00000000000011d3 in main at tarea.c:11
3       breakpoint     keep y   0x000000000000126c in main at tarea.c:18
4       breakpoint     keep y   0x0000000000001304 in main at tarea.c:24
(gdb) delete 3
(gdb) info break
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x00000000000011b5 in main at tarea.c:5
2       breakpoint     keep y   0x00000000000011d3 in main at tarea.c:11
4       breakpoint     keep y   0x0000000000001304 in main at tarea.c:24
(gdb)
```

Printing Variables and Expressions:

```
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break 18
Breakpoint 1 at 0x126c: file tarea.c, line 18.
(gdb) run
Starting program: /home/arindam/tarea

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Program to calculate the area of a triangle with integer coordinates
Enter the coordinates of the first corner: 3 8
Enter the coordinates of the second corner: 6 -7
Enter the coordinates of the third corner: -1 5

Breakpoint 1, main () at tarea.c:18
18              area = x1 * (y2 - y3);
(gdb) print x1
$1 = 3
(gdb) print x2
$2 = 6
(gdb) print x3
$3 = -1
(gdb) print x1+x2+x3
$4 = 8
(gdb) print x1+x2*x3
$5 = -3
(gdb) print x1*(y2-y3)+x2*(y1-y3)+x3*(y1-y2)
$6 = -33
(gdb) print $6/2
$7 = -16
(gdb)
```

Watching Variables after Every Step:

```
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break 16
Breakpoint 1 at 0x124d: file tarea.c, line 16.
(gdb) run
Starting program: /home/arindam/tarea

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Program to calculate the area of a triangle with integer coordinates
Enter the coordinates of the first corner: 3 8
Enter the coordinates of the second corner: 6 -7

Breakpoint 1, main () at tarea.c:16
16              scanf("%d%d", &x3, &y3);
(gdb) n
Enter the coordinates of the third corner: -1 5
18              area = x1 * (y2 - y3);
(gdb) display area
1: area = 4.59163468e-41
(gdb) n
19              area += x2 * (y1 - y3);
1: area = -36
(gdb) n
20              area += x3 * (y1 - y2);
1: area = -18
(gdb) n
21              if (area < 0) area = -area;
1: area = -33
(gdb) n
22                      area /= 2;
1: area = 33
(gdb) n
24              printf("Area of the triangle is %lf\n", area);
1: area = 16.5
(gdb) continue
Continuing.
Area of the triangle is 16.500000
[Inferior 1 (process 2711476) exited normally]
(gdb)
```

Setting Variables: The Program

Open *setvar.c

```c
#include <stdio.h>
#include <stdlib.h>

int main ()
{
        int a, b;

        a = 5;
        b = 8;
        printf("max(%d,%d) = %d\n", a, b, (a >= b) ? a : b);
        printf("max(%d,%d) = %d\n", a, b, (a >= b) ? a : b);
        exit(0);
}
```

Setting Variables: GDB

arindam@rahul-401: ~

```
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./setvar...
(gdb) break main
Breakpoint 1 at 0x1155: file setvar.c, line 8.
(gdb) run
Starting program: /home/arindam/setvar

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at setvar.c:8
8               a = 5;
(gdb) n
9               b = 8;
(gdb) n
10              printf("max(%d,%d) = %d\n", a, b, (a >= b) ? a : b);
(gdb) n
max(5,8) = 8
11              printf("max(%d,%d) = %d\n", a, b, (a >= b) ? a : b);
(gdb) set var b = 2
(gdb) print b
$1 = 2
(gdb) n
max(5,2) = 5
12              return 0;
(gdb) n
13      }
```

Conditions and Loops:

Open *average.c Save

```c
#include <stdio.h>
#include <stdlib.h>

int main ()
{
        int n, x, sum;

        n = sum = 0;
        printf("Keep on entering non-negative integers. Enter a negative integer to end.\n");
        while (1) {
        printf("Next number: "); scanf("%d", &x);
        if (x < 0) break;
        ++n; sum += x;
        }
        printf("Average of the numbers entered is %lf\n", (double)sum / (double)n);
        return 0;
}
```

Conditions and Loops: Example Run

```
(gdb) n
13              ++n; sum += x;
(gdb) n
11              printf("Next number: "); scanf("%d", &x);
(gdb) n
Next number: -1
12              if (x < 0) break;
(gdb) n
15              printf("Average of the numbers entered is %lf\n", (double)sum / (double)n);
(gdb) n
Average of the numbers entered is 5.666667
16              return 0;
(gdb) n
17      }
(gdb) n
Download failed: Invalid argument.  Continuing without source file ./csu/../sysdeps/nptl/libc_start_ca
__libc_start_call_main (main=main@entry=0x5555555551a9 <main>, argc=argc@entry=1,
    argv=argv@entry=0x7fffffffddf8) at ../sysdeps/nptl/libc_start_call_main.h:74
warning: 74     ../sysdeps/nptl/libc_start_call_main.h: No such file or directory
(gdb) n
[Inferior 1 (process 11222) exited normally]
(gdb)
```

```
Reading symbols from ./average...
(gdb) break main
Breakpoint 1 at 0x11b5: file average.c, line 5.
(gdb) run
Starting program: /home/arindam/average
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at average.c:5
5       {
(gdb) n
8               n = sum = 0;
(gdb) n
9               printf("Keep on entering non-negative integers. Enter a negative integer to end.\n");
(gdb) n
Keep on entering non-negative integers. Enter a negative integer to end.
11              printf("Next number: "); scanf("%d", &x);
(gdb) n
Next number: 5
12              if (x < 0) break;
(gdb) n
13              ++n; sum += x;
```

A Program with Non-Recursive Function Call:

Open tarea.c Save

```c
#include <stdio.h>
#include <stdlib.h>

double tarea ( int x1, int y1, int x2, int y2, int x3, int y3 )
{
        double area;

        area = x1 * (y2 - y3);
        area += x2 * (y1 - y3);
        area += x3 * (y1 - y2);
        if (area < 0) area = -area;
        area /= 2;
        return area;
}

int main ()
{
        int x1, y1, x2, y2, x3, y3;
        double area;

        printf("Program to calculate the area of a triangle with integer coordinates\n");
        printf("Enter the coordinates of the first corner: "); scanf("%d%d", &x1, &y1);
        printf("Enter the coordinates of the second corner: "); scanf("%d%d", &x2, &y2);
        printf("Enter the coordinates of the third corner: "); scanf("%d%d", &x3, &y3);
        area = tarea(x1,y1,x2,y2,x3,y3);
        printf("Area of the triangle is %lf\n", area);
        return 0;
}
```

Skip Going Inside The Function by Next:

```
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break main
Breakpoint 1 at 0x1266: file tarea.c, line 17.
(gdb) r
Starting program: /home/arindam/tarea
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
e[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at tarea.c:17
17      {
(gdb) next
21              printf("Program to calculate the area of a triangle with integer coordinates\n");
(gdb) next
Program to calculate the area of a triangle with integer coordinates
22              printf("Enter the coordinates of the first corner: "); scanf("%d%d", &x1, &y1);
(gdb) next
Enter the coordinates of the first corner: 3 8
23              printf("Enter the coordinates of the second corner: "); scanf("%d%d", &x2, &y2);
(gdb) next
Enter the coordinates of the second corner: 6 -7
24              printf("Enter the coordinates of the third corner: "); scanf("%d%d", &x3, &y3);
(gdb) next
Enter the coordinates of the third corner: -1 5
25              area = tarea(x1,y1,x2,y2,x3,y3);
(gdb) next
26              printf("Area of the triangle is %lf\n", area);
(gdb) next
Area of the triangle is 16.500000
27              return 0;
```

Step Into The Function:

```
Reading symbols from ./tarea...
(gdb) break main
Breakpoint 1 at 0x1266: file tarea.c, line 17.
(gdb) r
Starting program: /home/arindam/tarea
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at tarea.c:17
17      {
(gdb) next
21              printf("Program to calculate the area of a triangle with integer coordinates\n");
(gdb) next
Program to calculate the area of a triangle with integer coordinates
22              printf("Enter the coordinates of the first corner: "); scanf("%d%d", &x1, &y1);
(gdb) next
Enter the coordinates of the first corner: 3 8
23              printf("Enter the coordinates of the second corner: "); scanf("%d%d", &x2, &y2);
(gdb) next
Enter the coordinates of the second corner: 6 -7
24              printf("Enter the coordinates of the third corner: "); scanf("%d%d", &x3, &y3);
(gdb) next
Enter the coordinates of the third corner: -1 5
25              area = tarea(x1,y1,x2,y2,x3,y3);
(gdb) step
tarea (x1=3, y1=8, x2=6, y2=-7, x3=-1, y3=5) at tarea.c:8
8               area = x1 * (y2 - y3);
(gdb) next
9               area += x2 * (y1 - y3);
(gdb) print area
$1 = -36
(gdb) next
10              area += x3 * (y1 - y2);
(gdb) print area
$2 = -18
(gdb) finish
Run till exit from #0  tarea (x1=3, y1=8, x2=6, y2=-7, x3=-1, y3=5) at tarea.c:10
0x000055555555533d in main () at tarea.c:25
25              area = tarea(x1,y1,x2,y2,x3,y3);
Value returned is $3 = 16.5
(gdb) n
26              printf("Area of the triangle is %lf\n", area);
(gdb) n
Area of the triangle is 16.500000
27              return 0;
(gdb)
```

Forced Return From Function:

```
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./tarea...
(gdb) break main
Breakpoint 1 at 0x1266: file tarea.c, line 17.
(gdb) r
Starting program: /home/arindam/tarea
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at tarea.c:17
17      {
(gdb) n
21              printf("Program to calculate the area of a triangle with integer coordinates\n");
(gdb) n
Program to calculate the area of a triangle with integer coordinates
22              printf("Enter the coordinates of the first corner: "); scanf("%d%d", &x1, &y1);
(gdb) n
Enter the coordinates of the first corner: 3 8
23              printf("Enter the coordinates of the second corner: "); scanf("%d%d", &x2, &y2);
(gdb) n
Enter the coordinates of the second corner: 6 -7
24              printf("Enter the coordinates of the third corner: "); scanf("%d%d", &x3, &y3);
(gdb) n
Enter the coordinates of the third corner: -1 5
25              area = tarea(x1,y1,x2,y2,x3,y3);
(gdb) step
tarea (x1=3, y1=8, x2=6, y2=-7, x3=-1, y3=5) at tarea.c:8
8               area = x1 * (y2 - y3);
(gdb) n
9               area += x2 * (y1 - y3);
(gdb) print area
$1 = -36
(gdb) n
10              area += x3 * (y1 - y2);
(gdb) print area
$2 = -18
(gdb) return
Make tarea return now? (y or n) y
#0  0x000055555555533d in main () at tarea.c:25
25              area = tarea(x1,y1,x2,y2,x3,y3);
(gdb) n
26              printf("Area of the triangle is %lf\n", area);
(gdb) n
Area of the triangle is -18.000000
27              return 0;
(gdb)
```

Program With Recursive Function:

Open fib.c

```c
#include <stdio.h>
#include <stdlib.h>

int Fib ( int n )
{
        if (n < 0) return -1;
        if (n <= 1) return n;
        return Fib(n-1) + Fib(n-2);
}

int main ()
{
        int n, f;

        n = 8;
        f = Fib(n);
        printf("F(%d) = %d\n", n, f);
        return 0;
}
```

Program With Recursive Function: Line Numbers

```
arindam@rahul-401:~$ gdb ./fib
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./fib...
(gdb) l
1       #include <stdio.h>
2       #include <stdlib.h>
3
4       int Fib ( int n )
5       {
6               if (n < 0) return -1;
7               if (n <= 1) return n;
8               return Fib(n-1) + Fib(n-2);
9       }
10
(gdb) l
11      int main ()
12      {
13              int n, f;
14
15              n = 8;
16              f = Fib(n);
17              printf("F(%d) = %d\n", n, f);
18              return 0;
19      }
(gdb)
```

Function Call with Next:

```
arindam@rahul-401:~$ gdb ./fib
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./fib...
(gdb) break main
Breakpoint 1 at 0x11a1: file fib.c, line 15.
(gdb) r
Starting program: /home/arindam/fib
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at fib.c:15
15              n = 8;
(gdb) n
16              f = Fib(n);
(gdb) n
17              printf("F(%d) = %d\n", n, f);
(gdb) n
F(8) = 21
18              return 0;
(gdb) n
19      }
```

Function Call with Only One Step:

```
(gdb) break main
Breakpoint 1 at 0x11a1: file fib.c, line 15.
(gdb) r
Starting program: /home/arindam/fib
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at fib.c:15
warning: Source file is more recent than executable.
15              n = 8;
(gdb) n
16              f = Fib(n);
(gdb) step
Fib (n=8) at fib.c:6
6               if (n < 0) return -1;
(gdb) n
7               if (n <= 1) return n;
(gdb) n
8               return Fib(n-1) + Fib(n-2);
(gdb) n
9       }
(gdb) n
main () at fib.c:17
17              printf("F(%d) = %d\n", n, f);
(gdb) n
F(8) = 21
18              return 0;
(gdb) n
19      }
(gdb) continue
Continuing.
[Inferior 1 (process 63387) exited normally]
(gdb)
```

Enter Recursive Calls with Step:

```
(gdb) file ./fib
Load new symbol table from "./fib"? (y or n) y
Reading symbols from ./fib...
(gdb) break main
Breakpoint 2 at 0x11a1: file fib.c, line 15.
(gdb) r
Starting program: /home/arindam/fib
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, main () at fib.c:15
15              n = 8;
(gdb) n
16              f = Fib(n);
(gdb) s
Fib (n=8) at fib.c:6
6               if (n < 0) return -1;
(gdb) n
7               if (n <= 1) return n;
(gdb) n
8               return Fib(n-1) + Fib(n-2);
(gdb) n
9       }
(gdb) s
main () at fib.c:17
17              printf("F(%d) = %d\n", n, f);
(gdb) n
F(8) = 21
18              return 0;
(gdb) continue
Continuing.
[Inferior 1 (process 80000) exited normally]
(gdb)
```

Set Breakpoint with Function Name:

```
(gdb) file ./fib
Load new symbol table from "./fib"? (y or n) y
Reading symbols from ./fib...
(gdb) break main
Breakpoint 3 at 0x11a1: file fib.c, line 15.
(gdb) break Fib
Breakpoint 4 at 0x1159: file fib.c, line 6.
(gdb) r
Starting program: /home/arindam/fib
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 3, main () at fib.c:15
15              n = 8;
(gdb) n
16              f = Fib(n);
(gdb) n

Breakpoint 4, Fib (n=8) at fib.c:6
6               if (n < 0) return -1;
(gdb) n
7               if (n <= 1) return n;
(gdb) n
8               return Fib(n-1) + Fib(n-2);
(gdb) n

Breakpoint 4, Fib (n=7) at fib.c:6
6               if (n < 0) return -1;
(gdb) n
7               if (n <= 1) return n;
(gdb) n
8               return Fib(n-1) + Fib(n-2);
(gdb) n

Breakpoint 4, Fib (n=6) at fib.c:6
6               if (n < 0) return -1;
(gdb)
```

Stepping into An External Function:

```
Reading symbols from ./average...
(gdb) break main
Breakpoint 5 at 0x11b5: file average.c, line 5.
(gdb) r
Starting program: /home/arindam/average
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 5, main () at average.c:5
5       {
(gdb) s
8               n = sum = 0;
(gdb) s
9               printf("Keep on entering non-negative integers. Enter a negative integer to end.\n");
(gdb) s
Download failed: Invalid argument.  Continuing without source file ./libio/./libio/ioputs.c.
0x00007ffff7c87bf8 in __GI__IO_puts (str=0x555555556008 "Keep on entering non-negative integers. Enter a ne
warning: 35     ./libio/ioputs.c: No such file or directory
```

Program to Demonstrate The Call Stack:

Open *callstack.c

```c
#include <stdio.h>
#include <stdlib.h>

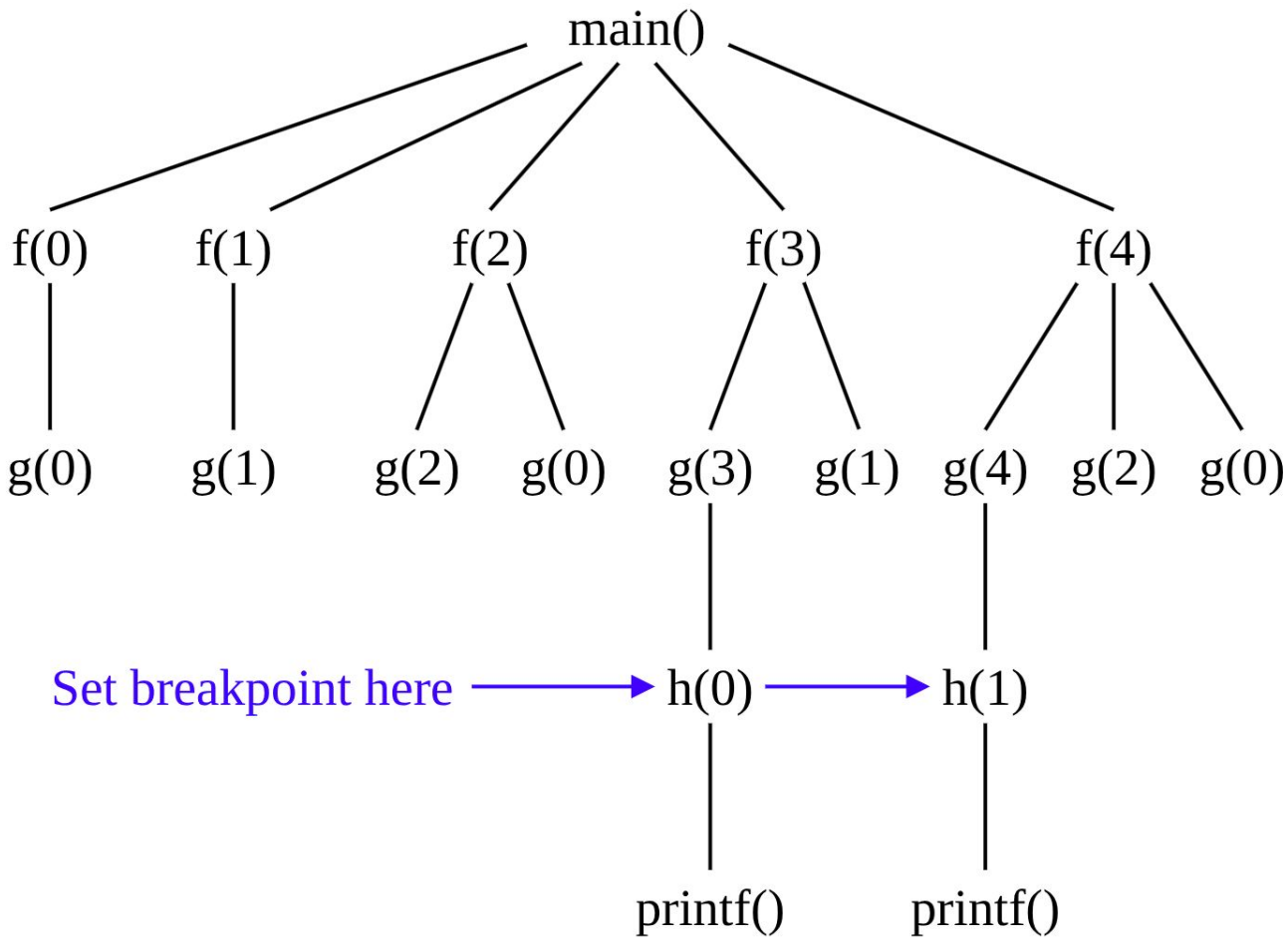
void h ( int n )
{
        printf("+++ Called h(%d)\n", n);
}

void g (int n )
{
        if (n >= 3) h(n-3);
}

void f ( int n )
{
        while (n >= 0) { g(n); n -= 2; }
}

int main ()
{
        int i;
        for (i=0; i<5; ++i) f(i);
        exit(0);
}
```

Call Tree:


main()
f(0)
f(1)
f(2)
f(3)
f(4)
g(0)
g(1)
g(2)
g(0)
g(3)
g(1)
g(4)
g(2)
g(0)
Set breakpoint here
h(0)
h(1)
printf()
printf()

View The Call Stack:

```
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./callstack...
(gdb) break h
Breakpoint 1 at 0x1178: file callstack.c, line 6.
(gdb) r
Starting program: /home/arindam/callstack
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, h (n=0) at callstack.c:6
6               printf("+++ Called h(%d)\n", n);
(gdb) backtrace
#0  h (n=0) at callstack.c:6
#1  0x00005555555551b6 in g (n=3) at callstack.c:11
#2  0x00005555555551d4 in f (n=3) at callstack.c:16
#3  0x0000555555555201 in main () at callstack.c:22
(gdb) step
Download failed: Invalid argument.  Continuing without source file ./stdio-common/./stdio-common/printf.c.
__printf (format=0x555555556004 "+++ Called h(%d)\n") at ./stdio-common/printf.c:32
warning: 32     ./stdio-common/printf.c: No such file or directory
(gdb) backtrace
#0  __printf (format=0x555555556004 "+++ Called h(%d)\n") at ./stdio-common/printf.c:32
#1  0x0000555555555191 in h (n=0) at callstack.c:6
#2  0x00005555555551b6 in g (n=3) at callstack.c:11
#3  0x00005555555551d4 in f (n=3) at callstack.c:16
#4  0x0000555555555201 in main () at callstack.c:22
(gdb) continue
Continuing.
+++ Called h(0)

Breakpoint 1, h (n=1) at callstack.c:6
6               printf("+++ Called h(%d)\n", n);
(gdb) backtrace
#0  h (n=1) at callstack.c:6
#1  0x00005555555551b6 in g (n=4) at callstack.c:11
#2  0x00005555555551d4 in f (n=4) at callstack.c:16
#3  0x0000555555555201 in main () at callstack.c:22
```

```
(gdb) up
#1  0x00005555555551b6 in g (n=4) at callstack.c:11
11              if (n >= 3) h(n-3);
(gdb) up
#2  0x00005555555551d4 in f (n=4) at callstack.c:16
16              while (n >= 0) { g(n); n -= 2; }
(gdb) info frame
Stack level 2, frame at 0x7fffffffdcb0:
 rip = 0x5555555551d4 in f (callstack.c:16); saved rip = 0x555555555201
 called by frame at 0x7fffffffdcd0, caller of frame at 0x7fffffffdc90
 source language c.
 Arglist at 0x7fffffffdca0, args: n=4
 Locals at 0x7fffffffdca0, Previous frame's sp is 0x7fffffffdcb0
 Saved registers:
  rbp at 0x7fffffffdca0, rip at 0x7fffffffdca8
(gdb) up
#3  0x0000555555555201 in main () at callstack.c:22
22              for (i=0; i<5; ++i) f(i);
(gdb) up
Initial frame selected; you cannot go up.
(gdb) down 2
#1  0x00005555555551b6 in g (n=4) at callstack.c:11
11              if (n >= 3) h(n-3);
(gdb) continue
Continuing.
+++ Called h(1)
[Inferior 1 (process 119439) exited normally]
(gdb)
```

View The Call Stack: Recursive Fibonacci Program

```
Breakpoint 1 at 0x1159: file fib.c, line 6.
(gdb) r
Starting program: /home/arindam/fib
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, Fib (n=8) at fib.c:6
warning: Source file is more recent than executable.
6               if (n < 0) return -1;
(gdb) continue 7
Will ignore next 6 crossings of breakpoint 1.  Continuing.

Breakpoint 1, Fib (n=1) at fib.c:6
6               if (n < 0) return -1;
(gdb) bt
#0  Fib (n=1) at fib.c:6
#1  0x000055555555517e in Fib (n=2) at fib.c:8
#2  0x000055555555517e in Fib (n=3) at fib.c:8
#3  0x000055555555517e in Fib (n=4) at fib.c:8
#4  0x000055555555517e in Fib (n=5) at fib.c:8
#5  0x000055555555517e in Fib (n=6) at fib.c:8
#6  0x000055555555517e in Fib (n=7) at fib.c:8
#7  0x000055555555517e in Fib (n=8) at fib.c:8
#8  0x00005555555551b2 in main () at fib.c:16
(gdb) c
Continuing.
```

```
Breakpoint 1, Fib (n=0) at fib.c:6
6               if (n < 0) return -1;
(gdb) bt
#0  Fib (n=0) at fib.c:6
#1  0x000055555555518d in Fib (n=2) at fib.c:8
#2  0x000055555555517e in Fib (n=3) at fib.c:8
#3  0x000055555555517e in Fib (n=4) at fib.c:8
#4  0x000055555555517e in Fib (n=5) at fib.c:8
#5  0x000055555555517e in Fib (n=6) at fib.c:8
#6  0x000055555555517e in Fib (n=7) at fib.c:8
#7  0x000055555555517e in Fib (n=8) at fib.c:8
#8  0x00005555555551b2 in main () at fib.c:16
(gdb) c
Continuing.

Breakpoint 1, Fib (n=1) at fib.c:6
6               if (n < 0) return -1;
(gdb) bt
#0  Fib (n=1) at fib.c:6
#1  0x000055555555518d in Fib (n=3) at fib.c:8
#2  0x000055555555517e in Fib (n=4) at fib.c:8
#3  0x000055555555517e in Fib (n=5) at fib.c:8
#4  0x000055555555517e in Fib (n=6) at fib.c:8
#5  0x000055555555517e in Fib (n=7) at fib.c:8
#6  0x000055555555517e in Fib (n=8) at fib.c:8
#7  0x00005555555551b2 in main () at fib.c:16
(gdb)
```



References:

1. https://kodekloud.com/courses/docker-for-the-absolute-beginner
2. https://docs.docker.com